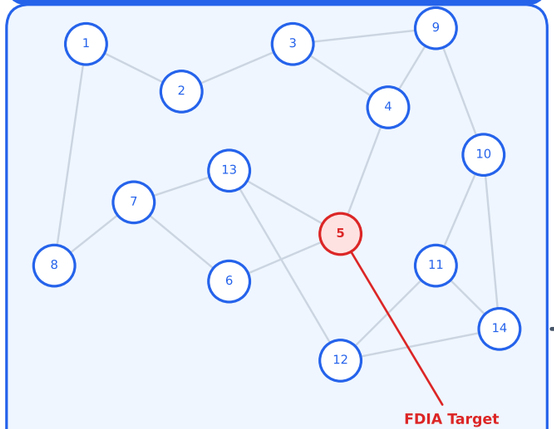
An Adaptive Explainable Graph Digital Twin for Cyber-Resilient Smart Grids

Physical Smart Grid / SCADA



Attack Vector: $a = h(x + c) - h(x)$

SCADA Readings: $[V, \theta, P, Q]^T$

Bad Data Filter: χ^2 BDD Bypassed ✗

SCADA RTUs • PMUs • Control Center

SCADA →

Virtual Graph Digital Twin

Graph Mirror: $G = (V, E, Y_{\text{bus}})$

Node Features: $x_i = [V_i, \theta_i, P_i, Q_i]^T$

Edge Attr: $e_{ij} = [R_{ij}, X_{ij}]^T$

GraphSAGE Layer 1: Aggregation

GraphSAGE Layer 2: Aggregation

GraphSAGE Layer 3: Aggregation

$$h_v^{(k)} = \text{ReLU}\left(W_k \cdot \left[h_v^{(k-1)} \| h_{\mathcal{N}(v)}^{(k)}\right]\right)$$

Loss: $\mathcal{L} = \mathcal{L}_{\text{BCE}} + 1.5\mathcal{L}_{\text{WCE}}$

Inductive Aggregation • Topology Invariant

Embeddings →

Multi-Task Detection & XAI

Detection Head (Graph-level)

Status: ATTACK DETECTED ✓

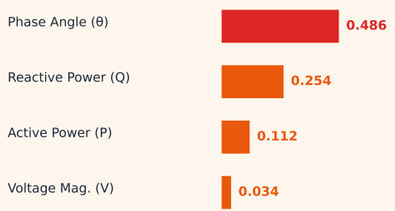
Det. F1: 98.25% ± 0.34% | AUC: 0.9993

Localization Head (Node-level)

Target: BUS 5 COMPROMISED ⚠

Loc. F1: 97.62% ± 0.84% | MCC: 0.982

IG Attribution (Bus 5 Anomaly):



Completeness: $\sum \text{Attr}_i = F(x) - F(x_0)$

Flag Bus 5 →

Closed-Loop Self-Healing

Step 1: Quarantine Bad RTU

Isolate Bus 5 Telemetry Channels

Step 2: Inject Baseline Forecast

$z_{\text{recon}} = z_{\text{nominal}} \pm \delta \quad (\delta \sim \pm 2\%)$

WLS State Estimator MSE:

Before Healing: 0.007228 p.u.²

After Healing: 0.000002 p.u.²

MSE Recovery: 99.97% | Latency < 2ms

Generalization Performance:

Outage Det. F1: 92.80% | Loc. F1: 86.96%

30-Bus Grid Transfer F1: 90.11%

Sub-millisecond Latency (0.908ms)

◄ Quarantine & Automated Healing Commands Sent to Field RTUs ◄